

LM555/NE555/SA555

Single Timer

Features

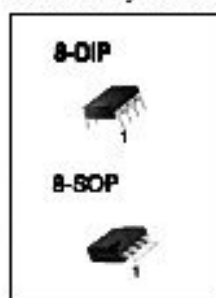
- High Current Drive Capability (200mA)
- Adjustable Duty Cycle
- Temperature Stability of 0.005%/°C
- Timing From μSec to Hours
- Turn off Time Less Than $2\mu\text{Sec}$

Applications

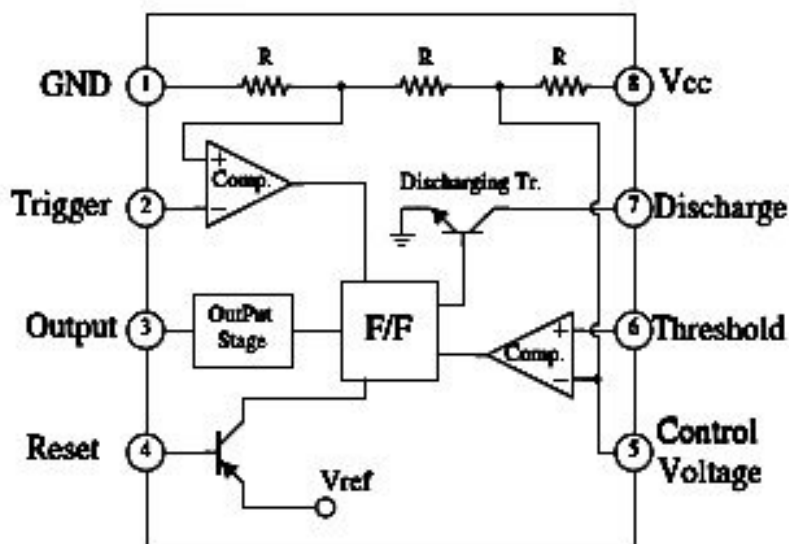
- Precision Timing
- Pulse Generation
- Time Delay Generation
- Sequential Timing

Description

The LM555/NE555/SA555 is a highly stable controller capable of producing accurate timing pulses. With a monostable operation, the time delay is controlled by one external resistor and one capacitor. With an astable operation, the frequency and duty cycle are accurately controlled by two external resistors and one capacitor.



Internal Block Diagram



Electrical Characteristics

($T_A = 25^\circ\text{C}$, $V_{CC} = 5 \sim 15\text{V}$, unless otherwise specified)

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Supply Voltage	V_{CC}	-	4.5	-	16	V
Supply Current (Low Stable) (Note1)	I_{CC}	$V_{CC} = 5\text{V}$, $R_L = \infty$	-	3	6	mA
		$V_{CC} = 15\text{V}$, $R_L = \infty$	-	7.5	15	mA
Timing Error (Monostable) Initial Accuracy (Note2) Drift with Temperature (Note4) Drift with Supply Voltage (Note4)	$ACCUR$ $\Delta t/\Delta T$ $\Delta t/\Delta V_{CC}$	$R_A = 1\text{k}\Omega$ to $100\text{k}\Omega$ $C = 0.1\mu\text{F}$	-	1.0 50 0.1	3.0 - 0.5	% ppm/ $^\circ\text{C}$ %/V
Timing Error (Astable) Initial Accuracy (Note2) Drift with Temperature (Note4) Drift with Supply Voltage (Note4)	$ACCUR$ $\Delta t/\Delta T$ $\Delta t/\Delta V_{CC}$	$R_A = 1\text{k}\Omega$ to $100\text{k}\Omega$ $C = 0.1\mu\text{F}$	-	2.25 150 0.3	- - -	% ppm/ $^\circ\text{C}$ %/V
Control Voltage	V_C	$V_{CC} = 15\text{V}$	9.0	10.0	11.0	V
		$V_{CC} = 5\text{V}$	2.6	3.33	4.0	V
Threshold Voltage	V_{TH}	$V_{CC} = 15\text{V}$	-	10.0	-	V
		$V_{CC} = 5\text{V}$	-	3.33	-	V
Threshold Current (Note3)	I_{TH}	-	-	0.1	0.25	μA
Trigger Voltage	V_{TR}	$V_{CC} = 5\text{V}$	1.1	1.67	2.2	V
		$V_{CC} = 15\text{V}$	4.5	5	5.6	V
Trigger Current	I_{TR}	$V_{TR} = 0\text{V}$	-	0.01	2.0	μA
Reset Voltage	V_{RST}	-	0.4	0.7	1.0	V
Reset Current	I_{RST}	-	-	0.1	0.4	mA
Low Output Voltage	V_{OL}	$V_{CC} = 15\text{V}$ $I_{SINK} = 10\text{mA}$ $I_{SINK} = 50\text{mA}$	-	0.06 0.3	0.25 0.75	V V
		$V_{CC} = 5\text{V}$ $I_{SINK} = 5\text{mA}$	-	0.05	0.35	V
High Output Voltage	V_{OH}	$V_{CC} = 15\text{V}$ $I_{SOURCE} = 200\text{mA}$ $I_{SOURCE} = 100\text{mA}$	12.75	12.5 13.3	-	V V
		$V_{CC} = 5\text{V}$ $I_{SOURCE} = 100\text{mA}$	2.75	3.3	-	V
Rise Time of Output (Note4)	t_R	-	-	100	-	ns
Fall Time of Output (Note4)	t_F	-	-	100	-	ns
Discharge Leakage Current	I_{LKG}	-	-	20	100	nA

Notes:

- When the output is high, the supply current is typically 1mA less than at $V_{CC} = 5\text{V}$.
- Tested at $V_{CC} = 5.0\text{V}$ and $V_{CC} = 15\text{V}$.
- This will determine the maximum value of $R_A + R_B$ for 15V operation, the max. total $R = 20\text{M}\Omega$, and for 5V operation, the max. total $R = 6.7\text{M}\Omega$.
- These parameters, although guaranteed, are not 100% tested in production.

Absolute Maximum Ratings (TA = 25°C)

Parameter	Symbol	Value	Unit
Supply Voltage	VCC	16	V
Lead Temperature (Soldering 10sec)	TLEAD	300	°C
Power Dissipation	PD	600	mW
Operating Temperature Range LM555/NE555 SA555	TOPR	0 ~ +70 -40 ~ +85	°C
Storage Temperature Range	TSTG	-65 ~ +150	°C